

Experiments :- Study of LAN Topology Network device and Address configuration.

Objective :-

- 1> To study the LAN topology used in college Campus.
- 2> To identify networking device & understand their function
- 3> To analyse wired & wireless communication method used in the network.
- 4> To measure speed of Network for wired & wireless connections.
- 5> To obtain MAC, IP Address & Subnet mask.

→ Problem Statement - 1

- o Draw & identify LAN topology of college campus & clearly label all components such as router, Switch & etc.

Solve :- The college campus network observed in this study consists of

- * Layer - 3 Core Switches in the server room.
- * Managed by PoE + Switches
- * Wireless Access point (APs)
- * Ethernet port in classroom & Laboratories

→ Network topology :-

②

Hybrid Hierarchical topology which combines :

- 1) Tree topology
- 2) Extended star topology
- 3) Ring topology (server room)

→ The network hierarchical layer

① Core layer

② Distribution layer

③ Access layer

① Core layer

* Core layer Contains server Room and server room have firewall, Router, 2 - L3 + switches, Patch Rack servers internet, etc.

② Distribution layer

* Distribution layer Contains 4 - switches

* The 4 switches are :-

① Main Build → R7 (Room no 213, 2nd floor)

② Main Build → R1 (Room no - 301, 3rd floor)

③ Main Build → R6 (Room no - 113, 1st floor)

④ Main Build → R2 (Room no - 201, 2nd floor)

* This 4 - Switches contains ③

① 18 port Gigabit Easy Smart Switch with 16 port. TL-SG1218MPE

② SG3428XMP → 24 port, Gigabit & 4-port 10 GE SFP+, L2 + Managed switches with 24 port PoE+.

→ Now Let's discuss these 2 ~~day~~ above Components :-

① TP-Link TL-SG1218MPE

* Likely used for : CCTV Camera wifi access point, etc.

* 18 port Gigabit "Easy Smart" Switch.

* 16 port support PoE+

② TP-Link Omada SG3428XMP

* Highest - end managed Layer 2 + switch

* 24 PoE+ port, uplinks, 10 GE SFP

* Usually acts as a main distribution for each floor

→ Access Layer

* Access Layer contain 3 - Switches i.e

1) Main Build R1 - [Room no - 301, 3rd floor]

2) Main Build R4 - [Room no - 05, Ground floor]

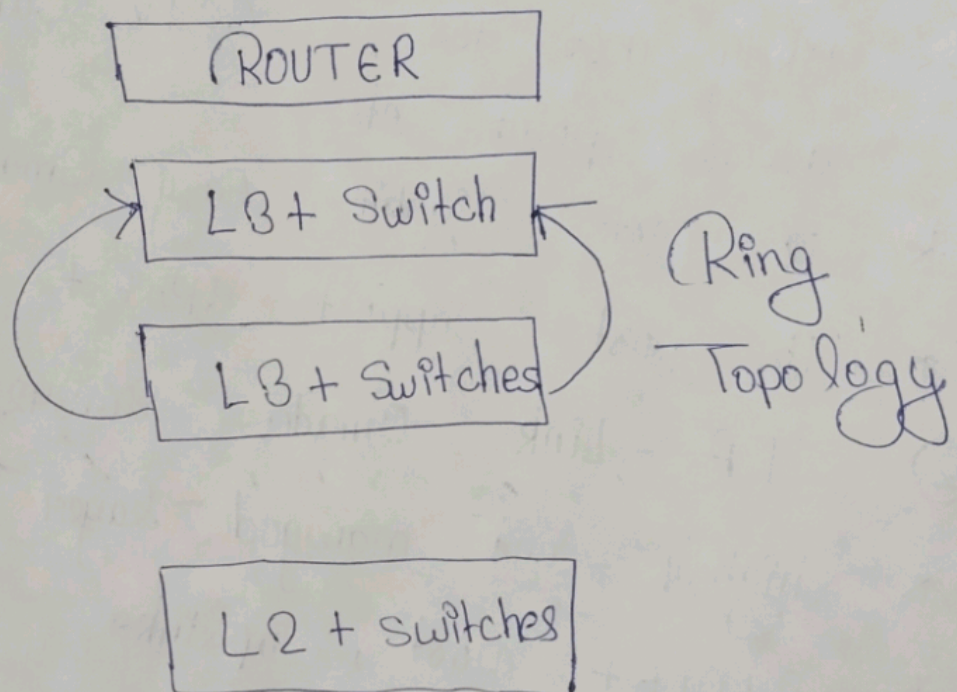
3) Main Build R5 - [Room no - 18, Ground floor]

* Every Switches act as access layer which helps to provide network connectivity to all Classroom, Lab, Faculty Room, etc (4)

* Access layer directly Connect

- (1) Ethernet port
- (2) Wireless Ap
- (3) Lab system
- (4) Outdoor wireless Ap (ex - 018)

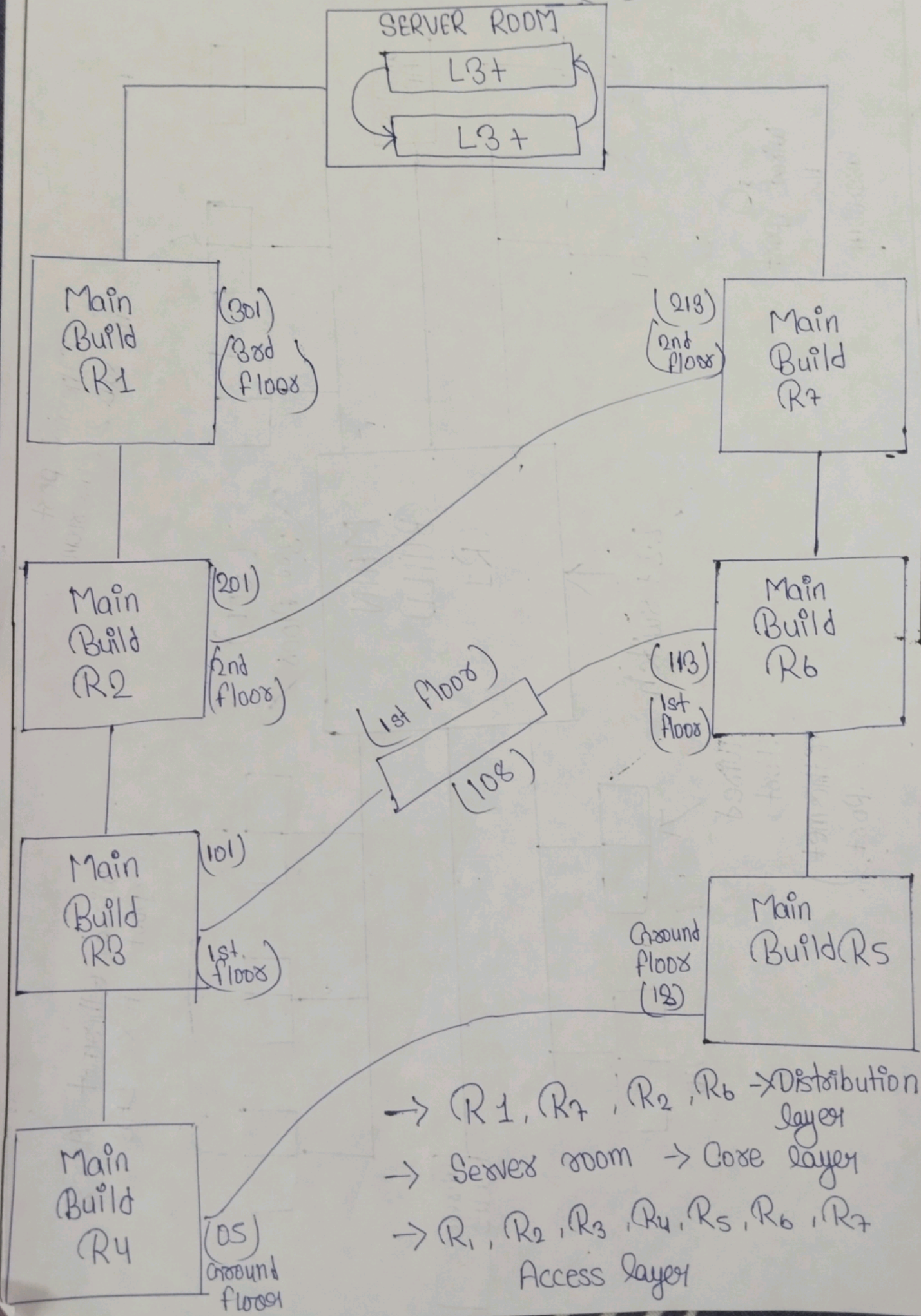
→ Server Room Architecture / topology



Topology

(5)

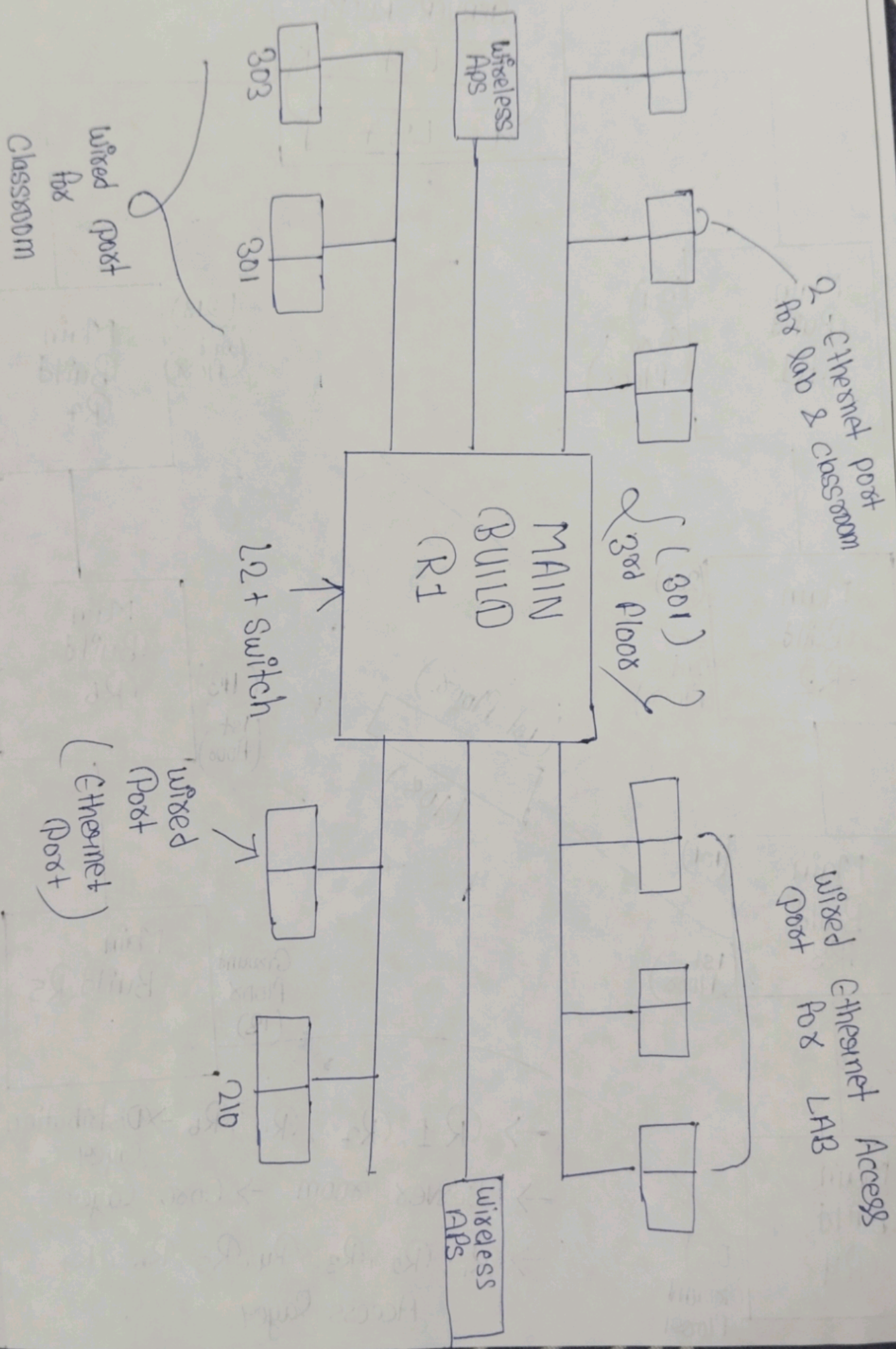
(210) 2nd floor



- R1, R7, R2, R6 → Distribution layer
- Server room → Core layer
- R1, R2, R3, R4, R5, R6, R7 Access layer

Access Layer Switch (3rd Floor - 301)

6



WIRELESS CONNECTION

```
C:\Users\conne>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . . : fe80::c880:b426:32f0:8dfd%20
    IPv4 Address. . . . . : 10.38.232.146
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.38.232.103

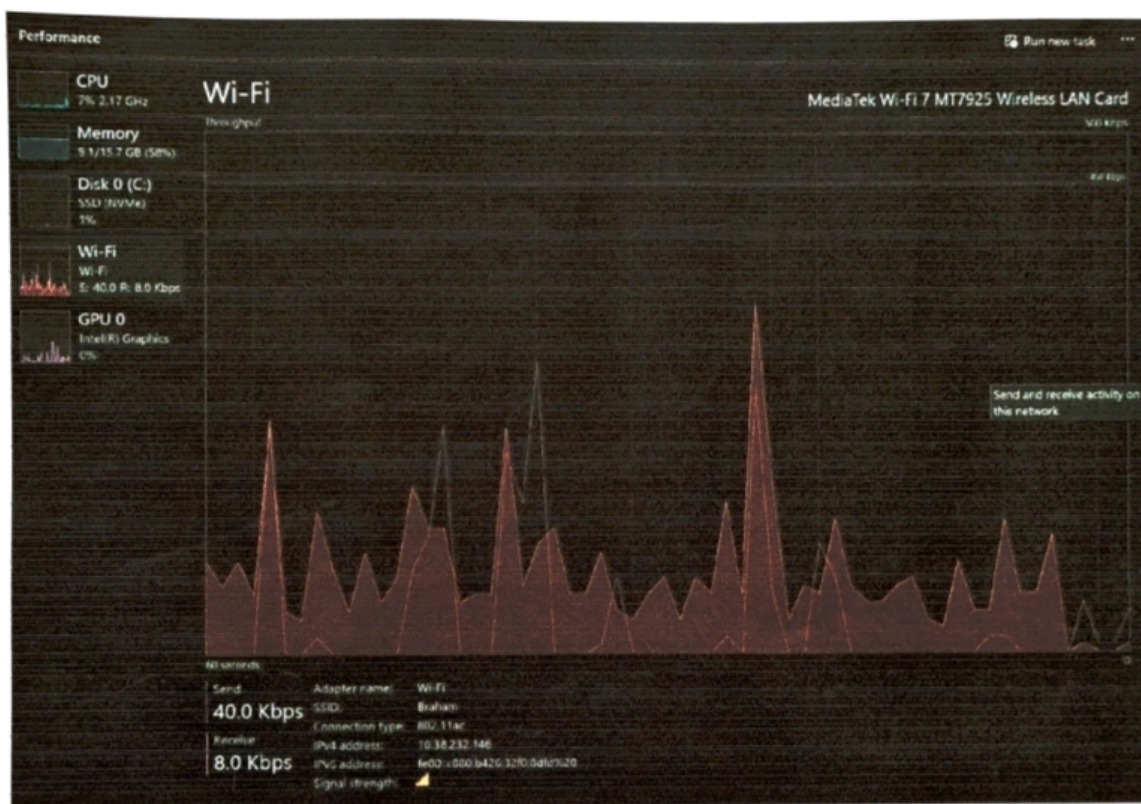
Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix . :
    IPv6 Address. . . . . : 2409:408a:1c8b:f98b:b142:f535:a2d7:cf02
    Temporary IPv6 Address. . . . . : 2409:408a:1c8b:f98b:d76:27d0:d0a4:5185
    Link-local IPv6 Address . . . . . : fe80::3c29:2fe7:1878:986d%9
    IPv4 Address. . . . . : 10.238.72.127
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::d8ed:73ff:fe9f:79de%9
                                10.238.72.213

C:\Users\conne>
```

↳ using "ipconfig"

WIRELESS SPEED



```
Command Prompt
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dell>getmac

Physical Address      Transport Name
=====
58-CD-C9-37-42-D3    \Device\NPF{676999331-FBF3-4F10-8349-303116AE3C5E}
8A-00-27-00-00-12    \Device\NPF{DB9A4153-556A-4E7B-A36E-D2915FB1DC3E}

C:\Users\dell>
```

→ MAC ADDRESS
→ Using "getmac"

WIRED CONNECTION

```
C:\Users\conne>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:408a:1c8b:f98b:b142:f535:a2d7:cf02
    Temporary IPv6 Address. . . . . : 2409:408a:1c8b:f98b:d76:27d0:d0a4:5185
    Link-local IPv6 Address . . . . . : fe80::3c29:2fe7:1878:986d%9
    IPv4 Address. . . . . : 10.238.72.127
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::d8ed:73ff:fe9f:79de%9
                                10.238.72.213

C:\Users\conne>
```

using IPCONFIG

→ Difference b/w wired & wireless speed? (7)

wired Network

- * Fastest Communication
- * Stable Connection
- * Low Latency
- * Dedicated Physical medium

Wireless Network

- * Slowest communication
- * Affected by distance / Interface
- * Higher Latency
- * Shared radio medium

→ Ethernet provide direct physical Communication using network cables, resulting in higher speed & stable performance.

→ Wireless Connections uses radio waves which are affected by interference, distance, obstacles & multiple connected users.

Q) Role of Subnet mask in network? (8)

→ A subnet mask identifies the network portion & host portion of an IP address. It helps device determines whether another device belongs to the same network or different network.

* It is important for :-

- ① Network Segment
- ② Efficient routing
- ③ Communication b/w devices

→ OBSERVATION

During the study of the college network infrastructure

- ① A centralized server room was used as network backbone.
- ② Two L3 switches were connected in redundant cyclic structure.
- ③ Managed TP-Link PoE+ switches were used in network cabinets.

→ Advantage of Existing Network ? (9)

- * Scalable architecture.
- * Centralized management.
- * Supports PoE+ device.
- * Easy Expansion of floors & classroom.
- * Support both wireless & wired
- * Redundant link for failover.

CONCLUSION

The Observed Campus LAN follow a Hybrid hierarchical topology consist of tree, Extended star, Ring. The network uses managed PoE+ switch, wireless access Point Ethernet port & centralized, server infrastructure to provide efficient wired & wireless Communication across the campus